

Probability & Statistics – The Backbone of AI/ML

Linear Algebra – The Backbone

TOPIC :

Problem solving on distribution

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① A random variable X has the Pdf
 $f(x) = ax(2-x)$, $0 \leq x \leq 2$
Find a , $E(x)$ and $\text{var}(2-3x)$

Solⁿ: Since $f(x)$ is density function.

$$\int_0^2 f(x) dx = 1$$

$$\Rightarrow \int_0^2 ax(2-x) dx = 1$$

$$\Rightarrow a \left[x^2 - \frac{x^3}{3} \right]_0^2 = 1$$

$$\Rightarrow a \left[4 - \frac{8}{3} \right] = 1 \Rightarrow a = \frac{3}{4}$$

Now,

$$\begin{aligned} E(x) &= \int_0^2 x \cdot \frac{3}{4} (2x - x^2) dx \\ &= \frac{3}{4} \left[\frac{2x^3}{3} - \frac{x^4}{4} \right]_0^2 \\ &= \frac{3}{4} \left[\frac{16}{3} - \frac{16}{4} \right] = \frac{3}{4} \times \frac{16}{12} = 1 \end{aligned}$$

$$\begin{aligned} E(x^2) &= \int_0^2 x^2 \cdot \frac{3}{4} x(2-x) dx \\ &= \frac{3}{4} \int_0^2 (2x^3 - x^4) dx \\ &= \frac{3}{4} \left[\frac{2x^4}{4} - \frac{x^5}{5} \right]_0^2 = \frac{3}{4} \left[8 - \frac{32}{5} \right] = \frac{3}{4} \times \frac{8}{5} = \frac{6}{5} \end{aligned}$$

$$\begin{aligned}\text{So, } \text{Var}(X) &= E(X^2) - \{E(X)\}^2 \\ &= \frac{6}{5} - (1)^2 \\ &= \frac{6}{5} - 1 = \frac{1}{5}\end{aligned}$$

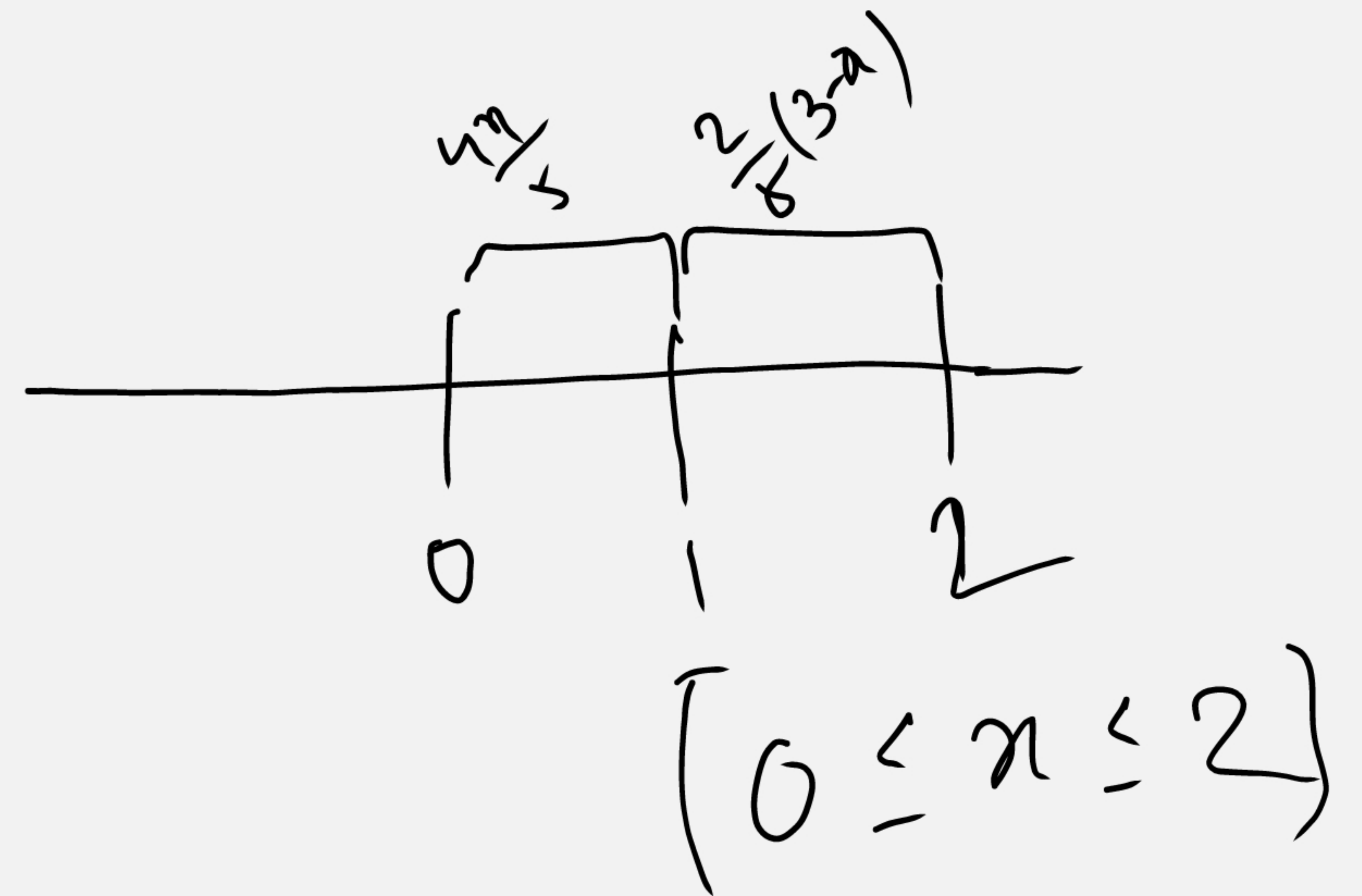
$$\begin{aligned}\text{Finally, } \text{Var}(2-3X) &= (-3)^2 \text{Var}(X) \\ &= 9 \times \frac{1}{5} = \frac{9}{5}\end{aligned}$$

Property :-

- (i) $\text{Var}(ax+b) = a^2 \text{Var}(X)$
- (ii) $\text{Var}(c) = 0$
- (iii) $\text{Var}(X) = E(X^2) - \{E(X)\}^2$
- (iv) $\text{S.D} = \sqrt{\text{Var}(X)}$

② Find $E(x)$ for the following density function:

$$f(x) = \frac{4x}{5}, 0 < x \leq 1 \\ = \frac{2}{5}(3-x), 1 < x \leq 2 \\ = 0, \text{ elsewhere}$$



Solⁿ:- $E(x) = \int_{-\infty}^{\infty} x f(x) dx$

$$= \int_0^1 x \cdot \frac{4x}{5} dx + \int_1^2 x \cdot \frac{2}{5}(3-x) dx$$
$$= \frac{4}{5} \left[\frac{x^3}{3} \right]_0^1 + \frac{2}{5} \left[\frac{3x^2}{2} - \frac{x^3}{3} \right]_1^2 = \frac{17}{15} \text{ Ans}$$

(3) Find the variance for the continuous random variable X with pdf.

$$f(x) = 1 - |1 - x|, 0 < x \leq 2$$

$$= 0, \text{ elsewhere}$$

$$\rightarrow f(x) \begin{cases} x & ; 0 < x \leq 1 \\ 2 - x & ; 1 < x \leq 2 \end{cases}$$

$$E(x) = \int_0^1 x \cdot x dx + \int_1^2 x \cdot (2 - x) dx$$

$$= \left[\frac{x^3}{3} \right]_0^1 + \left[\frac{2x^2}{2} - \frac{x^3}{3} \right]_1^2$$

$$= \frac{1}{3} + \left[4 - \frac{8}{3} - 1 + \frac{1}{3} \right] = 1$$

$$E(x^2) = \int_0^1 x^2 \cdot x dx + \int_1^2 x^2 \cdot (2 - x) dx$$

$$= \left[\frac{x^4}{4} \right]_0^1 + \left[\frac{2x^3}{3} - \frac{x^4}{4} \right]_1^2 = \frac{1}{4} + \frac{16}{3} - 4 - \frac{2}{3} + \frac{1}{4} = \frac{7}{6}$$

$$\frac{16+3-4-2}{12} = \frac{13}{12}$$

$$\text{Var} = E(x^2) - (E(x))^2$$

$$= \frac{7}{6} - (1)^2 = \frac{1}{6}$$

(0,1) (1,2) $\frac{|1-x|}{2}$

(0,2) $\text{Var} = E(x^2) - \{E(x)\}^2$

(0,1) (1,2) $E(x^2) = \int_0^1 x^2 \cdot x dx + \int_1^2 x^2 \cdot (2 - x) dx$

④ For what value of a will the function
 $f(x) = ax; x = 1, 2, 3, \dots, n$
be the probability mass function of a discrete
random variable X ?
Find the mean and variance of X ?

(H/w) Problem